

•[31]: #12412485

```
[2]: import numpy as np

np.random.seed(42)

population = np.random.exponential(scale=5, size=100000)

print("Population Mean:", np.mean(population))

Population Mean: 4.979850795280925
```

```
[3]: sample_size = 1000
num_samples = 20

sample_means = []

for i in range(num_samples):
    sample = np.random.choice(population, size=sample_size, replace=True)
    sample_means.append(np.mean(sample))
```

```
[4]: print("Average of Sample Means:", np.mean(sample_means))

Average of Sample Means: 5.03481913553641
```

```
[5]: sample_means = np.array(sample_means)
```

```
[6]: print("\nPopulation Mean:", np.mean(population))
print("Average of Sample Means:", np.mean(sample_means))
```

```
Population Mean: 4.979850795280925
Average of Sample Means: 5.03481913553641
```

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